

# TFPT — Research Contracts for the two open gates

$(U_{\text{wall}})$ : the parabolic flavor wall-selection    and     $(G_{\text{metric}})$ : the full quantum-gravity measure

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## What this document is about

After the compiler closure the entire residual is

$$\text{Rest} = (U_{\text{wall}}) \oplus (G_{\text{metric}}) \oplus (F_{\text{frontier}})$$

This note turns the two genuine research gates into *contracts*: a numbered chain of lemmas, the single theorem that closes each gate, and — for every step — whether it is machine-certifiable today.  $(F_{\text{frontier}})$  (Koide,  $\eta_B$ , axion relic scale,  $m_p/m_e$ ) is *not* a gate: those are deliberately typed QFT/cosmology interfaces (see `tfpt_4_frontier`). **Priority:**  $(U_{\text{wall}})$  *first* (finite, algebraic, falsifiable), then  $(G_{\text{metric}})$  (deep analytic programme).

**Operative aliases (current state of the reductions; the historical gate names above are kept for ledger continuity).** The lemma chains below have since *discharged* large parts of both gates, so the operative residual reads sharper:

$$\text{Rest}_{\text{operativ}} = v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$$

—  $(U_{\text{wall}}) \rightarrow v_{\text{geo}}$ : the quark *ratios* are closed combinatorially (Readout Rigidity + Plücker, U2/v49/v71),  $U_{\text{point}}$  collapses to the *one* dimensionful unit  $v_{\text{geo}}$  shared with  $1/G$  (v75/v68) — what remains is a scale anchor, not a flavor gap (the *full* amplitude matrix beyond the ratios is the only place where  $U_{\text{point}}$  still appears);  $(G_{\text{metric}}) \rightarrow G_{\text{net}}$ : holographically reduced to the single boundary-net statement “*the seam-Calderón inclusion has Jones index 4 =  $|\mu_4|$* ” — holomorphy,  $(E_8)_1$  and the unique 2D bulk then *follow* (v89/v92/GATE.METRIC.05-07); and the carrier-side QBL input now merges into the *same* statement: the tower carrier  $\rightarrow SO(16)_1 \rightarrow (E_8)_1$  carries one and the same 16-fermion content, whose single quasi-free kernel (rank =  $c$ ; Wick/Pfaffian exact) is the certified Calderón datum — closing  $G_{\text{net}}$  therefore also closes the carrier choice [L] ([`verification/v113_quasifree_kernel.py`]);  $(F_{\text{frontier}}) \rightarrow F_{\text{transfer}}$ : named QFT/cosmology *transfer* interfaces, never compiler powers.

## Research Contract 1 — $(U_{\text{wall}})$

**Goal.** Prove

$$\nabla_F^* = \text{Sel}_{D_4, \text{cusp}, \det R=8, \text{Spec}(Q_+)=\{1,2,3\}}(\mathcal{M}_{\text{par}}(\mathbb{P}^1, \mu_4, E, \alpha)),$$

with  $E = \mathcal{O}(-2) \oplus \mathcal{O}(-1) \oplus \mathcal{O}(-1)$  and  $\alpha = \{0, \frac{1}{3}, \frac{2}{3}\}$  at the four points of  $\mu_4$ . The substrate is standard: degree-zero stable parabolic bundles correspond to unitary representations with prescribed local monodromies (Mehta–Seshadri); tame non-abelian Hodge gives the Higgs / flat-connection side. The TFPT content is to pin the *one*  $D_4$ -symmetric, rank-3, four-point point that realises the selectors.

**Lemma 1** (U0 — normalisation). *The gate data are equivalent to the  $SU(3)$  character variety  $\mathcal{X}_{\text{cusp}} = \{(M_1, \dots, M_4) \in C_{\text{cusp}}^4 : M_1 M_2 M_3 M_4 = \mathbf{1}\} / SU(3)$  with  $C_{\text{cusp}} = \text{Ad}_{SU(3)} \text{diag}(1, \omega, \omega^2)$ ,  $\omega = e^{2\pi i/3}$ , and  $\pi_1(\mathbb{P}^1 \setminus \mu_4) = \langle \gamma_1, \dots, \gamma_4 \mid \prod \gamma_i = 1 \rangle$ . To show:  $\mathcal{M}_{\text{par}}^{D_4} \simeq \mathcal{X}_{\text{cusp}}^{D_4}$  as polystable*

*Mehta–Seshadri / tame Hodge realisation.* Machine: the group presentation and  $D_4$ -action are finite and codable (Sage/Wolfram).

**Lemma 2** (U1 — wall landscape, corrected). *Parabolic degree zero forces the line weight-sums onto the wall  $(w_1, w_2, w_3) = (2, 1, 1)$ . An explicit balanced representative (weights in  $\frac{1}{3}$ ) is*

$$W_{\text{wall}} = \frac{1}{3} \begin{pmatrix} 2 & 1 & 2 & 1 \\ 1 & 0 & 0 & 2 \\ 0 & 2 & 1 & 0 \end{pmatrix},$$

*each column a permutation of  $(0, 1, 2)$ , row sums  $(2, 1, 1)$ . **Honest result:** the  $6^4=1296$  column-permutation matrices give 144 walls; under the full group  $D_4 \times S_3(\text{lines}) \times S_3(\text{values})$  they form five orbits, not one. So  $W_{\text{wall}}$  is not singled out by symmetry — the uniqueness must come from the selector (Lemma U4), not from a symmetry quotient. The wall landscape is nonetheless a finite, explicitly enumerated set;  $W_{\text{wall}}$  is one of the five. Machine: fully certified  $C_U^{(1)}$  ([verification/v29\_research\_contract\_certs.py]).*

**The splitting type is now algebraic (RH route, [verification/v32\_rh\_splitting.py]).** *Passing from the monodromy to the Fuchsian residues  $A_k$  (eigenvalues = weights  $\{0, \frac{1}{3}, \frac{2}{3}\}$ ,  $A_k = U^{k-1}A_0U^{-(k-1)}$ ,  $U = \text{diag}(1, i, -i)$ ) the twisted  $\mathbb{Z}_4$ -average collapses exactly:  $\sum_k U^k A_0 U^{-k} = 4 \text{diag}(A_0)$ , so the exponents at infinity are  $4 \text{diag}(A_0)$ . A flat bundle with regular  $\infty$  needs integer exponents summing to  $4 = -\deg E$ ; with the cusp weights the only options are perms of  $(2, 1, 1)$  and  $(2, 2, 0)$ , i.e.*

$$\boxed{\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2 \iff \text{diag}(A_0) = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})}$$

*(the sibling  $(2, 2, 0) \rightarrow \mathcal{O}(0) \oplus \mathcal{O}(-2)^2$ ). Such a Hermitian  $A_0$  (spectrum  $\{0, \frac{1}{3}, \frac{2}{3}\}$ , that diagonal) exists by Schur–Horn. So the splitting is an algebraic condition on  $\text{diag}(A_0)$ . Honest wall: the global  $\prod M_k = \mathbf{1}$  is the path-ordered monodromy (not  $\exp(2\pi i \sum A_k)$ ), constraining off-diag( $A_0$ ) via the genuine RH solve; and  $R$ -extraction still needs the  $C_6 \leftrightarrow \mathbb{P}^1 \setminus \mu_4$  bridge. The RH route fixes the splitting algebraically but not  $c_u/c_d$ .*

**Lemma 3** (U2 — the kill switch, run and hardened [N]). *The selected polystable wall representative must not collapse to the diagonal abelian  $U(1)^3$  point if it is to give non-trivial  $SU(3)_F$  transport mixing: (A)  $\nabla_F^*$  polystable but not simultaneously diagonal (the desired breakthrough), or (B) diagonal collapse, in which case  $c_u/c_d$  stays honestly [P]. **Result: the cheap test is run robustly and lands on A.** The kill-switch fires iff the  $D_4$ -symmetric cusp representation is reducible (commutant non-scalar). Across 400 random cusp-class generators of the  $\mathbb{Z}_4$  family  $M_k = U^{k-1}MU^{-(k-1)}$  the commutant is scalar at every point (irreducible 400/400), the reducible locus is hit with probability 0 (measure-zero, codimension  $\geq 1$ ), and the mixing strength stays bounded away from zero (min off-diag = 0.35 > 0.05). So case A is the generic behaviour, not an accident of the one explicit point of [verification/v33\_explicit\_flat\_bundle.py]: ( $U_{\text{wall}}$ ) survives its own cheapest falsification test. [verification/v38\_uwall\_killswitch.py]*

**Lemma 4** (U3 —  $D_4$  fixed locus as an explicit variety, positive-dimensional). *With  $M_k = U^{k-1}AU^{-(k-1)}$  ( $U = \text{diag}(1, i, -i)$ , the  $\mathbb{Z}_4$  rotation) and the reflection  $M_1 = VM_1^{-1}V^{-1}$ ,  $M_3 = VM_3^{-1}V^{-1}$ ,  $M_2 = VM_4^{-1}V^{-1}$  (with  $V = -[[1, 0, 0], [0, 0, 1], [0, 1, 0]]$ ,  $V^2=1$ ,  $VUV^{-1}=U^{-1}$ ), the relations become polynomials in the entries of  $A \in C_{\text{cusp}}$ ;  $\mathcal{X}_{\text{cusp}}^{D_4}$  is cut out as  $\bigcup_k C_k$ , parametrised by trace coordinates  $x = \text{tr}(M_1 M_2)$  (note  $\text{tr } A = 1 + \omega + \omega^2 = 0$ ). **Result** ( $C_U^{(2)}$ , [verification/v30\_d4\_character\_variety.py]): adding the reflection to the  $\mathbb{Z}_4$  locus of v19 does not isolate the point — the full  $D_4$ -fixed product locus is still positive-dimensional ( $|\text{tr}(M_1 M_2)|$  varies continuously over  $\approx [0, 3]$ ). So even the full  $D_4$  symmetry cannot select  $\nabla_F^*$ . Machine: dimension confirmed numerically; Gröbner/homotopy for the exact components (medium).*

**Lemma 5** (U4 — selectors as equations). *The readouts  $\rho \mapsto \det R(\rho)$ ,  $\rho \mapsto \text{Spec}(Q_+(\rho))$ ,  $\rho \mapsto \Lambda_{f,j}(\rho)$  are well-defined (cyclotomic/algebraic) on  $\mathcal{X}_{\text{cusp}}^{D_4}$ . **Acceptance criterion:***

$$\#\{\rho \in \mathcal{X}_{\text{cusp}}^{D_4} : \det R = 8, \text{Spec}(Q_+) = \{1, 2, 3\}\} = 1$$

in the polystable quotient. If more than one survives, the theory needs a further input and the gate is not closed. **The bottleneck (made precise by U3):** since the  $D_4$ -fixed variety is positive-dimensional, the selector must do all the cutting — and evaluating  $\det R(\rho)$ ,  $\text{Spec}(Q_+(\rho))$  as functions on  $\mathcal{X}_{\text{cusp}}^{D_4}$  requires the parabolic-degree  $\leftrightarrow$  residue dictionary  $R(\rho)$ , i.e. the H2 equivalence. That dictionary is the single remaining analytic input of the ( $U_{\text{wall}}$ ) gate.

**What  $R(\rho)$  is** ([verification/v31\_R\_dictionary.py]). Two facts pin its nature. (i) Case A is alive: every  $D_4$ -fixed cusp monodromy found is irreducible, so the wall point carries non-trivial  $SU(3)_F$  mixing — the U2 kill switch does not fire to the diagonal case B. (ii)  $R(\rho)$  is not algebraic:  $R$  is integer-valued, but the algebraic invariants of  $\rho$  (e.g.  $\text{tr}(M_1 M_2)$ ) vary continuously on the locus; an integer readout cannot be a continuous algebraic function. Hence  $R(\rho)$  is the discrete invariant of the non-abelian Hodge / Mehta–Seshadri map (the parabolic-degree / Hodge-filtration jumps of the harmonic bundle  $E_\rho$ ): integer output, but its computation is the transcendental harmonic-metric (Hitchin) solve on the generic Higgs locus. **Refinement ( $U_4''$ ,** [verification/v40\_harmonic\_metric.py]): at the polystable point TFPT actually selects this pessimism does not apply — see the box below.

### Progress: the selector side of U4 is closed on the explicit point [N]

While  $R(\rho)$  as a function on the whole locus needs the Hitchin solve, the two U4 selectors are *algebraically* accessible on the explicit flat bundle of [verification/v33\_explicit\_flat\_bundle.py] — no PDE. Computed exactly from  $A_0$ : the exponents at infinity  $\text{eig}(\sum_k A_k) = \{2, 1, 1\}$  give the splitting type  $E = \mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$ , i.e. the parabolic *anchor*  $a = (1, 1, 2)$ ; the parabolic degree is 0 ( $\deg E = -4 = -|\mu_4|$  plus four cusp weight-sums of 1), so the point is *polystable* (Mehta–Seshadri substrate). The two selectors then read off the bundle directly:

$$\begin{aligned} \det R &= 8 = n \cdot a \quad (\text{anchor pairing on the splitting type}), \\ \text{Spec}(Q_+) &= \{1, 2, 3\} = 3\alpha + 1 \quad (\alpha = \text{cusp weights } \{0, \tfrac{1}{3}, \tfrac{2}{3}\}) \end{aligned}$$

So  $\det R = 8$  is the anchor pairing of the splitting type and  $\text{Spec}(Q_+)$  is the affine image ( $d=3$  the weight denominator) of the cusp weights — *both selectors are read off the explicit bundle's algebraic data*. **Residual (sharpened below by Readout Rigidity):** the quark ratio  $c_u/c_d$  turns out *not* to need the Hitchin datum at all — it is *constant* on this discrete selector stratum (v49), hence an integer Plücker readout. The only genuinely transcendental input left is the *absolute* amplitude normalisation ( $U_{\text{point}}$ ), which is an anchor, not a missing PDE. [verification/v39\_uwall\_selectors.py]

### Breakthrough: the harmonic metric is *finite linear algebra*, not a PDE [N]

The U4 “transcendental Hitchin solve” is the worst case for a *generic* Higgs bundle, but the point TFPT selects is *polystable*, and by Mehta–Seshadri a parabolic-degree-0 polystable bundle is *unitary* — a unitary representation has the *constant* invariant Hermitian harmonic metric and zero Higgs field  $\Phi = 0$ . So at the selected point the harmonic metric is finite linear algebra. On the explicit bundle of [verification/v33\_explicit\_flat\_bundle.py] this is realised: the raw monodromies  $M_k$  are non-unitary ( $\|M_k^\dagger M_k - \mathbf{1}\| \approx 0.83$ ), but there is a *unique* common invariant Hermitian form  $H$  ( $\dim \ker\{M_k^\dagger H M_k = H\} = 1$ ), it is *positive-definite*, and the unitarised holonomy  $\widetilde{M}_k = H^{1/2} M_k H^{-1/2}$  is unitary ( $\sim 10^{-8}$ ), lies in the cusp class  $\{1, \omega, \omega^2\}$  with  $\prod \widetilde{M}_k = \mathbf{1}$ , and has *clean* matrix elements  $|\widetilde{M}_k|_{ij} \in \{0, \frac{1}{2}, \frac{1}{\sqrt{2}}\}$ . **So  $C_U^{(3)}$  reduces from a transcendental PDE to: this finite unitarisation (done) + the combinatorial word-length  $R$  (closed: H1 +  $\det R = 8$ ).** *Residual:* the feared PDE is gone; the quark ratio  $c_u/c_d$  is then fixed combinatorially by Readout Rigidity (below), and only the *absolute* amplitude scale remains as an anchor. [verification/v40\_harmonic\_metric.py]

### The final leg-assignment test (honest result) [N]

Does the harmonic-frame holonomy  $\{0, \frac{1}{2}, \frac{1}{\sqrt{2}}\}$  feed the  $\delta = \frac{1}{2}$  resolvent to give the lepton amplitudes  $\Lambda = (0.475, 1.107, 0.917)$ ? **(i) Ruled out as a literal identity:**  $\Lambda_\mu = 1.107 > 1$ , while unitary holonomy entries have modulus  $\leq 1$ , so the amplitudes *cannot* be holonomy matrix elements — they are the non-unitary resolvent Green function (as v19 already noted). **(ii) Suggestive positive:** the harmonic-frame holonomy diagonal modulus is exactly  $|\text{diag } \widetilde{M}_0| = (0, \frac{1}{2}, \frac{1}{2})$ , and that  $\frac{1}{2}$  *equals* the distinguished lepton transport value  $\delta = \frac{1}{2}$  that v20 used by hand — so the geometry plausibly *supplies*  $\delta$  (an observation at the explicit point; genericity across the locus is future work). **(iii) Clean negative:** no natural holonomy construction (e.g.  $(\mathbf{1} - \frac{1}{2}\widetilde{M})^{-1}$ ) maps the holonomy to the amplitude vector. **Verdict:** the amplitudes are the  $\delta = \frac{1}{2}$  resolvent (closed combinatorially via the word-length  $R$ ); the holonomy supplies  $\delta$  and the leg selection, not the amplitude magnitude.  $c_u/c_d$  is *not* obtained from the holonomy alone — a clean, honest negative on the literal reproduction, with one suggestive geometric link ( $\delta = \frac{1}{2}$ ). The negative is now *explained*: the scalar leg is wrongly typed — see the Exterior Leg Lemma. [verification/v41\_leg\_assignment.py]

### The torsion normal form and the $\delta = \frac{1}{2}$ theorem [I]/[N] ([verification/v114\_torsion\_delta.py])

The v41 “genericity is future work” item is now executed — and yields a theorem. *(i) Flatness*  $= \mu_4$  *torsion* [I]: for any  $M$  (symbolic),  $\prod_k U^k M U^{-k} = (MU)^4$  — the  $\mathbb{Z}_4$ -family flatness constraint *is* the torsion statement “ $T = MU$  is a fourth root of unity”: the  $\mu_4$  atom appears as the *order* of the twisted cusp generator. *(ii) The  $\delta = \frac{1}{2}$  theorem* [I] (both directions): on the *involutive* branch ( $T^2 = \mathbf{1}$ )  $T$  is a Hermitian reflection  $2vv^\dagger - \mathbf{1}$ , and the cusp trace condition  $2v^\dagger U^{-1}v = 1$  splits *exactly* into  $|v_1|^2 = \frac{1}{2}$  (real part) and  $|v_2| = |v_3|$  (imaginary part), forcing

$$\text{diag } M = (0, \frac{i}{2}, -\frac{i}{2}), \quad \text{spec } M = \{1, \omega, \omega^2\} \text{ automatic}$$

( $\chi_M = \lambda^3 - 1$  from unitary + trace 0 + det 1). So  $\delta = |M_{22}| = \frac{1}{2}$  holds *exactly and constantly* on the whole two-parameter branch — the distinguished lepton transport value that v20 used by hand is a  $\mu_4$ -torsion *identity*, not an observation. *(iii) Reflection lemma* [I]: the  $D_4$  reflection forces  $\text{diag } M = (r, z, \bar{z})$  with  $r$  real and  $\text{Re } z = -r/2$  — the whole  $D_4$  diagonal freedom is *one* real parameter, and  $\delta = \frac{1}{2} \iff r = 0$ . *(iv) Branch census* [N] (seeded): the full  $D_4$ -fixed flat cusp locus realises only  $\text{tr}(MU) \in \{1, -1\}$ ; the involutive branch has the diagonal *constant*  $(0, \frac{1}{2}, \frac{1}{2})$  to machine precision, while on the  $\{1, i, -i\}$  branch  $d_1$  varies over  $[0, 1]$  — and the explicit v33/v40 point sits exactly in its  $d_1 = 0$  slice, the *same*  $\delta$  value. *Residue* [P] (recorded): whether the anchor splitting  $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$  (Mehta–Seshadri side) forces the  $d_1 = 0$  slice — or selects the involutive branch — is the remaining question;  $\delta = \frac{1}{2}$  itself is no longer the open item.

### The anchor pins the residue: an exact normal form for the $(U_{\text{wall}})$ bundle [I]/[N] ([verification/v115\_anchor\_residue.py])

The v114 branch question is answered — and the v33 *numerical* Riemann–Hilbert point acquires an *exact closed form*. *(i)  $\mu_4$ -average lemma* [I]:  $\sum_k U^k X U^{-k} = 4 \text{diag } X$  for any  $X$  —  $\mu_4$  conjugation-averaging *is* the diagonal projection, so the exponents at infinity are  $4 \text{diag } A_0$  and the anchor splitting holds *iff*  $\text{diag } A_0 = (2, 1, 1)/4 = a/|\mu_4|$ : the residue diagonal **is** the anchor over  $\mu_4$  (v33’s diagonal was forced, not chosen). *(ii) The  $(8, 0, 5)/144$  lemma* [I]: the anchor diagonal and the cusp spectrum  $\{0, 1, 2\}/N_{\text{fam}}$  pin  $\sum |a_{ij}|^2 = \frac{13}{144}$ , and with  $a_{13} = 0$

the determinant condition solves *uniquely* to

$$\boxed{(|a_{12}|^2, |a_{13}|^2, |a_{23}|^2) = \frac{(8, 0, 5)}{144}} \quad \begin{array}{l} 8=\text{rank } E_8, \ 5=g_{\text{car}}, \\ 144=(|\mu_4|N_{\text{fam}})^2, \ 8+5=13=\Delta_Q \end{array}$$

— the off-diagonal weights are compiler atoms, and the total 13 is the *same*  $\Delta_Q$  that denominates  $c_u/c_d = 55/117 = 5 \cdot 11/(9 \cdot 13)$ . (iii) *Exact normal form [I]*:  $A_0^* = \begin{bmatrix} 1/2 & \sqrt{2}/6 & 0 \\ \sqrt{2}/6 & 1/4 & \sqrt{5}/12 \\ 0 & \sqrt{5}/12 & 1/4 \end{bmatrix}$  has  $\chi = \lambda(\lambda - \frac{1}{3})(\lambda - \frac{2}{3})$  *exactly*. (iv)  $A_0^*$  is the RH solution [N]: its big-circle monodromy is trivial to  $10^{-10}$ ,  $\prod M_k = \mathbf{1}$ , and the unitarised holonomy has  $|\text{diag } \widetilde{M}_0| = (0, \frac{1}{2}, \frac{1}{2})$ ,  $\text{tr}(\widetilde{M}_0 U) = 1$  — the v33/v40 point is gauge-equivalent to  $A_0^*$  (the  $a_{12}, a_{23}$  phases are diagonal gauge). (v) *The anchor forces the  $d_1 = 0$  slice [N]*: a multi-seed RH scan over the anchor-pinned spectral manifold lands at *every* flat solution on the *same* gauge orbit with  $|\text{diag } \widetilde{M}_0| = (0, \frac{1}{2}, \frac{1}{2})$  — combined with v114, the whole harmonic diagonal is **anchor + torsion forced**;  $\delta = \frac{1}{2}$  needs no selection. *Residue [P]*:  $a_{13} = 0$  and the gauge-orbit uniqueness are numerical (one component across all seeds); promoting them to [I]/[L] via Gröbner on the RH constraints is the remaining  $U_{\text{point}}$  step.

**The resonance theorem:**  $M_\infty = \mathbf{1} \iff a_{13} = 0$  — one gauge orbit [I]  
([verification/v116\_resonance\_uniqueness.py])

The anticipated Gröbner promotion turned out to be a *linear* computation. Writing the equivariant system at infinity as  $dY/dw = -(1/w)(B_0 + B_1 w + \dots)Y$  with  $B_m = \sum_k i^{km} U^k A_0 U^{-k}$ , the *twisted*  $\mu_4$  averages collapse [I]:

$$B_0 = 4 \text{diag } A_0, \quad \boxed{B_1 = 4a_{12}E_{12} + 4\overline{a_{13}}E_{31}}$$

(only the eigenvalue-ratio  $-i$  cells survive). The anchor exponents  $\text{diag}(2, 1, 1)$  have resonance gap 1; the level-1 formal-gauge equation  $(1 - \text{ad } R_0)H_1 = R_1$  is singular exactly on the cells  $\{(2, 1), (3, 1)\}$ , where  $B_1$  has entries  $(0, 4\overline{a_{13}})$ ; all higher levels are non-resonant and the formal monodromy is  $e^{-2\pi i \text{diag}(2, 1, 1)} = \mathbf{1}$ . Hence [I]:

$$\boxed{M_\infty = \mathbf{1} \iff a_{13} = 0}$$

— the transcendental Riemann–Hilbert condition collapses to *one linear equation*. *Uniqueness corollary [I]*:  $a_{13} = 0$  + the  $(8, 0, 5)/144$  moduli + phases = diagonal gauge  $\Rightarrow$  the flat anchor locus is **exactly one gauge orbit**, the exact  $A_0^*$ : within the  $\mu_4$ -equivariant anchor class the  $(U_{\text{wall}})$  bundle datum is *algebraically unique*. *Sharpness [N]*:  $\|M_\infty - \mathbf{1}\| \sim 8.5 |a_{13}|$  (0.17 at  $a_{13} = 0.02$ ) vs  $10^{-10}$  at  $A_0^*$ . *Honest scope [P]*: the equivariant ansatz is the established  $D_4$  selector input (v19/v30); the *residue* side is now [I], while the holonomy values (the harmonic diagonal  $(0, \frac{1}{2}, \frac{1}{2})$ ) remain [N] — transcendental monodromy of an exact system.

**The monodromy is the Weyl group of  $A_3$  — the holonomy is exact [I]/[N]**  
([verification/v117\_monodromy\_weyl\_a3.py])

The last [N] item of the chain — the holonomy *values* — closes: the unitarised monodromy of the exact  $A_0^*$  system is itself *exact*, with entries in  $\frac{1}{2}\mathbb{Z}[i]$ :

$$\boxed{\widetilde{M}_0 = \begin{pmatrix} 0 & -\frac{1+i}{2} & \frac{1-i}{2} \\ -\frac{1+i}{2} & -\frac{i}{2} & -\frac{1}{2} \\ \frac{1-i}{2} & -\frac{1}{2} & \frac{i}{2} \end{pmatrix}}$$



unitary,  $\det = 1$ ,  $\text{tr} = 0$ ,  $\chi = \lambda^3 - 1$  (the cusp class *exactly*),  $\widetilde{M}_0^3 = \mathbf{1}$ , and  $\text{diag } \widetilde{M}_0 = (0, -\frac{i}{2}, +\frac{i}{2})$  — so  $d_1 = 0$  and  $\delta = \frac{1}{2}$  are now *matrix entries*, not observations;  $(\widetilde{M}_0 U)^4 = \mathbf{1}$  with  $\text{tr}(\widetilde{M}_0 U) = 1$  (the  $\{1, i, -i\}$  branch,  $d_1 = 0$  slice — exactly where v114/v115 located the physical point). *The group [I]*: exact enumeration of  $\langle U, \widetilde{M}_0 \rangle$  gives order 24 with order statistics  $(1, 9, 8, 6)$  and character values  $(3, -1, 0, 1)$  — uniquely  $S_4$  in its standard  $\otimes$  sign representation, i.e. the (twisted) reflection representation of the **Weyl group of the family lattice**  $A_3$ :

$$\text{monodromy}(U_{\text{wall}}) = W(A_3) = S_4, \quad |W(A_3)| = 4! = 24$$

— the flavor wall realises the  $A_3$  factor of the glue  $D_5 \oplus A_3 + \mu_4$  as its Weyl group ( $U =$  a 4-cycle, the  $\mu_4$  deck;  $\widetilde{M}_0$  a 3-cycle, the family rotation). *Identification [N]*: the ODE monodromy of  $A_0^*$  equals the exact  $\widetilde{M}_0$  to  $3.5 \times 10^{-10}$ , and the invariant form  $H$  is  $U$ -invariant to  $10^{-11}$ . *Status*: holonomy values [I]; only the analytic identification stays [N]. The  $\delta$  thread (v20  $\rightarrow$  v40/v41  $\rightarrow$  v114  $\rightarrow$  v115  $\rightarrow$  v117) is closed end to end.

## The hexagon is the sign-twisted family spectrum — first exact H2 piece [I] ([verification/v118\_hexagon\_family\_dictionary.py])

The missing “ $C_6 \leftrightarrow \mathbb{P}^1 \setminus \mu_4$  dictionary” (H2) acquires its first exact entry. *Sign-twist lemma [I]*:  $-\widetilde{M}_0$  has order 6 and

$$\text{spec}(\widetilde{M}_0) \cup \text{spec}(-\widetilde{M}_0) = \mu_6$$

— the six-site hypercharge hexagon spectrum *is* the family monodromy spectrum together with its sheet twist ( $\mathbb{Z}_6 = \mathbb{Z}_2 \times \mathbb{Z}_3$  realised as  $(-1) \times \langle \widetilde{M}_0 \rangle$ ); the v20 denominators  $\frac{5}{4} - \cos(r\pi/3)$  are exactly the eigen-denominators  $|1 - \zeta_6^r/2|^2$  of the two family resolvents  $(\mathbf{1} \mp \frac{1}{2}\widetilde{M}_0)^{-1}$ . *Cyclotomic determinant [I]*:  $\det(\mathbf{1} - t\widetilde{M}_0) = 1 - t^3 = (1 - t)\Phi_3(t)$ , so  $\det(\mathbf{1} \mp \frac{1}{2}\widetilde{M}_0) = (\frac{7}{8}, \frac{9}{8})$  — and the lepton coefficients become resolvent determinants:  $c_e = |\mu_4|/(2 \det_-) = \frac{16}{7}$ ,  $c_\mu = N_{\text{fam}}/(2 \det_+) = \frac{4}{3}$ ,  $c_\tau = |\mu_4| \det_- / N_{\text{fam}} = \frac{7}{6}$ , product rule  $= |\mu_4|/\det_+ = \frac{32}{9}$ . The lepton 7 and 9 *are* the family-resolvent determinants;  $\delta = \frac{1}{2} = |(\widetilde{M}_0)_{22}|$  is supplied by the matrix itself. *Sheet-extended group [I] (audit)*:  $\langle U, \widetilde{M}_0, -\mathbf{1} \rangle$  has order  $48 = \Omega_{\text{adm}} = N_{\text{fam}} \dim S^+$  — the seed’s quartic coefficient. *Residue [P]*: the dictionary’s *values* are exact; its *address table* (which hexagon site each fermion occupies; the quark composition) remains v20’s established input / open — nothing re-fished.

## The address table: the lepton words are the compiler atoms [I]/[P] ([verification/v120\_address\_table.py])

The address table acquires its structure — on word-length integers *frozen* in v18/v20, long before today’s readings (the red-team firewall). (i) *The lepton words are the atoms [I]*:

$$(L_e, L_\mu, L_\tau) = (8, 5, 3) = (\text{rank } E_8, g_{\text{car}}, N_{\text{fam}}) = (p_0 + e_2, e_2, p_0)$$

in mass order (longest word = lightest lepton). (ii) *Addresses = division by the hexagon [I]*:  $(r, w) = L \text{ divmod } p_2$  with  $p_2 = 6 = |R^+(A_3)|$  itself an atom:  $e \rightarrow (2, 1)$ ,  $\mu \rightarrow (5, 0)$ ,  $\tau \rightarrow (3, 0)$ ;  $\sum r = 10 = p_3 = \mathcal{A}_\Lambda$ ,  $\sum L = 16 = \dim S^+$ . (iii) *Sheet parity [I]*: by the v118 dictionary,  $e$  sits on the *untwisted* sheet ( $\omega \in \text{spec } \widetilde{M}_0$ ),  $\mu, \tau$  on the *twisted* sheet  $(-\omega, -1)$ ; the anchor lepton  $\tau$  occupies the unique *real* twisted eigenvalue  $-1$  — exactly the site where v20’s product rule replaces the resolvent. (iv) *Quark sum rules [I] (frozen v18 words 7, 7, 5, 3, 2, 0)*: up-type sum  $= 10 = p_3$ ; down-type sum  $= 14 = p_1 + p_3$ ; **all quarks**  $= 24 = |W(A_3)|$

(the monodromy group order, v117); **all nine charged fermions** =  $40 = p_1 p_3 = a^T R a$  (v119); quark  $\sum r = 2p_2$ ,  $\sum w = |\mathbb{Z}_2|$ ; the top sits at the vacuum site  $(0, 0)$  — zero word, no suppression, the established ladder anchor. *Residue [P]*: five independent closures  $(16, 10, 14, 24, 40)$  constrain the address map; the per-fermion *assignment* remains the v18/v20 input — deriving it is the remaining H2 content.

**The address table is pinned:  $R$  is the unique hexagon matrix with atom margins [I]** ([verification/v121\_address\_pinning.py])

The per-fermion assignment closes at the *information* level. The established identity  $L = R + 2U$  (v95) says the word-length table is  $L[\text{sector}][\text{generation}]$  — rows (up; down; lepton) =  $((7, 3, 0); (7, 5, 2); (8, 5, 3))$ , the v18/v20 words verbatim — so the addresses are the entries of the *residue operator*  $R$  plus one hexagon turn for the first generation, and  $R$ 's first column is the **anchor**  $a = (1, 1, 2)$ . The margins are atoms [I]: row sums  $(4, 8, 10) = (e_1, \text{rank } E_8, p_3)$ , column sums  $(4, 13, 5) = (e_1, \Delta_Q, g_{\text{car}})$ . And the census closes it [I]:

among all  $3 \times 3$  matrices with entries in the hexagon  $\{0..5\}$ ,  
exactly 17 have the atom margins — and exactly ONE has  $\det = 8$

— and it is  $R$  (also the unique one with SNF =  $(1, 1, 8)$ ; all 17 determinants are distinct, 8 occurs once — control computed explicitly). **The assignment table carries zero residual information beyond the atom margins and the rank determinant.** Within-sector ordering is the established transport reading ( $m \sim \lambda_Y^L$ : longer word = lighter fermion). *Residue [P]*: the open H2 item shrinks from “derive 9 per-fermion integers” to “derive the six atom margins +  $\det = \text{rank}$ ”. *Firewall*:  $R$ ,  $L$  and all word lengths were frozen in v4/v18/v20/v71 long before this census — no degrees of freedom spent.

**The margins are theorems** ([verification/v122\_margin\_theorem.py]). Even the margins are not input. Three selectors, all established and frozen *before* the address question was posed —  $(S1)$  the  $D_4$  annihilator  $n = (5, -9, 6)$  with  $n^T R = (8, 0, 0)$  (it kills the unchanged  $(c_2, c_3)$  plane;  $n \cdot a = 8$ ),  $(S2)$   $\det R = 8$ ,  $(S3)$   $\det K = 4$  with  $K = R + Q\Sigma$  — pin  $R$  *uniquely* among hexagon matrices [I]: the  $(S1)$  census has  $4 \times 4 \times 4 = 64$  candidates,  $(S2)$  leaves 12,  $(S3)$  leaves **exactly one**, and it is  $R$ . Hence the atom margins, the anchor column  $Re_1 = a$  and all sum rules above are *consequences* — their input status is revoked. *Bonus lemma [I]*: on the  $(S1) + (S2)$  locus  $\det(M + 2U) = 20 = \det L$  holds for *all* 12 candidates — the fourth quartet determinant is redundant, the compiler's familiar overdetermination. *Residue [P]*: H2 reduces to the establishment of the selectors themselves ( $n$  and the diamond determinants — pre-existing inventoried objects); the address question introduces *no new residual class*.

**Exterior Leg Lemma — the quark  $u/d$  leg is a  $\Lambda^2 F$  area, not a scalar [U]/[P]**

The v41 negative is a *category error*, not a dead end.  $u$  and  $d$  share the hexagon-radial data  $(r, w) = (1, 1)$  ( $L = 7$  for both), and the  $C_6$  resolvent  $c(r, w) = |\mu_4|^w / (\frac{5}{4} - \cos \frac{r\pi}{3})$  reads *only*  $(r, w)$  — so any *scalar* leg gives  $c_u/c_d \equiv 1$ . A scalar sees radius, not orientation. The  $u/d$  separation lives in the smallest *non-scalar* invariant of the  $SU(3)_F$  composition: the exterior square  $\Lambda^2 F$ , i.e. the oriented anchor-plane area  $\text{Pl}_{1,a}(K)$ . The “11” is then *generated*, not read:

$$(K, \mathbf{1}, a) \rightarrow B_K = \begin{pmatrix} 13 & 8 & 4 \\ 18 & 11 & 6 \end{pmatrix} \rightarrow \text{Pl}(K) = (-1, 6, 4) = (-N_\Phi, |R^+(A_3)|, |\mu_4|) \rightarrow \|\text{Pl}(K)\|_1 = 11,$$

$$\frac{c_u}{c_d} = \frac{g_{\text{car}} \|\text{Pl}_{1,a}(K)\|_1}{N_{\text{fam}}^2 \Delta_Q} = \frac{5 \cdot 11}{9 \cdot 13} = \frac{55}{117} \quad (\text{anchor microcode} = \frac{e_2(a)(p_3(a)+1)}{p_0(a)^2(2p_2(a)+1)}).$$

**Typing:** [I] for the Plücker / microcode identities ( $\text{Pl}(K) = (-1, 6, 4)$ ,  $\|\cdot\|_1 = 11$ ,  $5 \cdot 11 / (9 \cdot 13) = 55/117$ ); and — by Readout Rigidity (v49, below) — [I] also for the *physical* ratio  $C_{ud}(\rho^*) = \text{this } \Lambda^2 F \text{ readout}$ , which is *constant* on the derived selector stratum and so does *not* need any point-selection on  $\Lambda^2 F$ . The quark residual is thereby re-typed from “find the right scalar  $y$ ” to a closed exterior readout for the *ratio*; only the *absolute* amplitude scale ( $U_{\text{point}}$ ) stays as an anchor, no new scalar, no fishing.

*Audit (the 121 lemma, [verification/v119\_review\_validation\_2.py]):* the “11” has a second, independent appearance [I]: with  $h = \mathbf{1} + a = (2, 2, 3)$  in the canonical anchor ordering,  $h^T R h = 121 = 11^2 = \|\text{Pl}(K)\|_1^2 = (p_3 + 1)^2$  — the quadratic *anchor norm* of the residue operator  $R$  reproduces the Plücker integer (sensitivity control: the three orderings of  $h$  give  $\{105, 121, 135\}$ ; only the canonical one yields 121). Bonus checksums:  $\mathbf{1}^T R \mathbf{1} = 22 = 2 \times 11$  (the entry sum of  $R$ ) and  $a^T R a = 40 = p_1 p_3 = 10 b_1 - 1$ . Typed *audit*, not load-bearing — the oriented Plücker mechanism above remains the derivation. [verification/v42\_exterior\_leg.py]

**Bridge test** ([verification/v43\_exterior\_bridge.py]). The typing is confirmed: for the  $SU(3)$  holonomy the exterior square is the conjugate representation,  $\Lambda^2(\widetilde{M}) = \widetilde{M}$  exactly — the exterior leg *is* the  $\bar{\mathbf{3}}$ . But the *continuous* action  $\Lambda^2(\widetilde{M}) \cdot (\mathbf{1} \wedge a)$  does *not* equal the *integer*  $\text{Pl}(K) = (-1, 6, 4)$ : the integer Plücker is the combinatorial  $K$  datum. So the bridge  $\rho^* \rightarrow \text{Pl}(K)$  is the *discrete* non-abelian-Hodge invariant (the integer  $R$ ;  $\det R = 8$  and  $\text{Spec}(Q_+) = \{1, 2, 3\}$  already confirmed on the bundle, Lemma U4 box), not a continuous exterior-square computation — exactly the H2 equivalence, now pinned from the  $\Lambda^2 F$  side.

**Lie-level grounding — no special math** ([verification/v44\_carrier\_exterior.py]). The discrete invariant has a home already in TFPT: the carrier half-spinor is the *even exterior algebra* of the 5-slot carrier,  $16 = \Lambda^{\text{even}}(5) = \binom{5}{0} + \binom{5}{2} + \binom{5}{4} = 1 + 10 + 5$ , and under  $SU(5) \subset SO(10)$  the up quark sits in  $10 = \Lambda^2(5)$  while the down sits in  $\bar{5} = \Lambda^4(5)$ . So the *exterior degrees* are  $\deg(u^c) = 2$ ,  $\deg(d^c) = 4$ , with  $\deg(u) + \deg(d) = 6 = |R^+(A_3)|$  (the hexagon) and  $\deg(d) - \deg(u) = 2 = |\mathbb{Z}_2|$  (the sheet). The “ $\Lambda^2$ ” of the exterior leg is therefore not exotic Hodge data — it is the carrier’s own  $\Lambda^2$  grading (the up quark *lives* in  $\Lambda^2(5)$ ). This re-types the residual once more: from “discrete non-abelian-Hodge invariant” to “the carrier exterior degree” — a standard [L] fact. *Honest scope:* this grounds the *type*; the Plücker *number* 55/117 stays the family  $\text{Pl}(K)$  [I], and the carrier- $\Lambda^2 \leftrightarrow$  family- $\Lambda^2 F$  link (via family  $\otimes$  carrier = 15 = 16–1) plus the normalisation remain [P].

**The “11” is Pascal — a third, simplest origin** ([verification/v45\_family\_exterior.py]). Following that link,  $\text{Pl}(K)$  is the *exterior algebra of the family fundamental*  $4 = \mu_4$  (the  $A_3 = SU(4)$  fundamental):  $\Lambda^k(4) = \binom{4}{k}$  is Pascal row 4 = (1, 4, 6, 4, 1) with full sum  $2^4 = 16 = \dim S^+$ , and  $|\text{Pl}(K)| = (1, 4, 6) = \{\Lambda^0, \Lambda^1, \Lambda^2\}(4)$  are exactly its distinct exterior powers. Hence

$$\|\text{Pl}(K)\|_1 = \sum_{k=0}^2 \binom{4}{k} = 11 = 2^4 - \binom{4}{3} - \binom{4}{4} = 16 - g_{\text{car}}$$

(the cumulative  $\Lambda^\bullet(\mu_4)$  up to degree 2 =  $\deg(u^c)$ ), and family  $\otimes$  carrier = 15 =  $\dim \mathfrak{su}(4) = \Lambda^2(5) \oplus \Lambda^4(5) = 10 + 5 = N_{\text{fam}} g_{\text{car}}$  read three ways. So the “11” is generated a *third* way — the cumulative family exterior algebra up to the up-quark’s degree — pure Pascal/branching, *no special math*. The [P] physical normalisation is unchanged; the algebraic origin of 55/117 is now triply grounded.

**Lemma 6** (U5 — the “11” worry, *discharged* by Readout Rigidity). *The original worry was that  $c_u/c_d = C_{ud}(\rho^*)$  must be evaluated at the uniquely selected  $\rho^*$  without using 11 as input, so that until then 55/117 would be a mere table-level rational shadow. This is now resolved (v49):  $C_{ud}$  is constant on the discrete selector stratum  $\mathcal{S}_{8,Q}$ , so the “11” is earned as the rigid Plücker readout  $\|\text{Pl}(K)\|_1$  (v42) — independently of point-uniqueness. Point-uniqueness of  $\rho^*$  enters only the absolute amplitude scale, not the ratio.*



**Lemma 7** (U6 — the four certificates).  $C_U^{(1)}$  wall enumeration (done);  $C_U^{(2)}$   $D_4$ -fixed character solver;  $C_U^{(3)}$  selector uniqueness (one  $S$ -equivalence class);  $C_U^{(4)}$  amplitude extraction — the ratios  $R, Q, c_u/c_d$  are read off the (derived) selector stratum (closed,  $v49/v71$ ); only the absolute  $U_f^*, c_q$  normalisation is the anchor.

### Progress: an explicit valid flat bundle is realised (RH solve) [N]

A numerical Riemann–Hilbert solve produced an *explicit* Hermitian residue  $A_0$  (eigenvalues  $\{0, \frac{1}{3}, \frac{2}{3}\}$ ,  $\text{diag} = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$ ) for which the Fuchsian system  $A(z) = \sum_k U^{k-1} A_0 U^{-(k-1)} / (z - i^k)$  simultaneously satisfies: the cusp class at each puncture; the splitting  $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$  (exponents  $\sum_k A_k = \text{diag}(2, 1, 1)$ ); the *path-ordered* monodromy at infinity is trivial,  $\|M_\infty - \mathbf{1}\| \sim 10^{-9}$ , i.e.  $M_1 M_2 M_3 M_4 = \mathbf{1}$ ; and the representation is *irreducible* — **case A** (non-trivial  $SU(3)_F$  mixing, not the diagonal case B). So a valid ( $U_{\text{wall}}$ ) flat bundle *exists and is constructed*; the U2 kill switch lands on A. *Honest scope*: this is one realising point (existence + case A), likely one of a residual family; the unique  $\nabla_F^*$  still needs the  $\det R = 8$  selector and  $c_u/c_d$  still needs the  $C_6 \leftrightarrow \mathbb{P}^1 \setminus \mu_4$  dictionary (H2). [verification/v33\_explicit\_flat\_bundle.py]

*H2 bridge probed* ([verification/v34\_h2\_bridge\_attempt.py]). The explicit per-puncture monodromies  $M_k$  (cusp class,  $\prod M_k = \mathbf{1}$  to  $10^{-5}$ ) were computed; but the natural resolvent-dressed diagonal extraction ( $|\text{diag } M_k| = (0, \frac{1}{2}, \frac{1}{2})$  times the  $C_6$  resolvent factor) does *not* reproduce the lepton amplitudes (0.475, 1.107, 0.917). So the precise  $\Gamma_{ij}^{\min}$  geodesic-to-word dictionary (how the 6-site hypercharge hexagon combines with the 3-dim family  $\rho^*$ ) is the genuine remaining analytic input — it is *not* fixed by a natural guess, and we do *not* fish for 55/117.

### Theorem U — the four-way split (sharper than one monster gate) [F]/[A]

$D_4$  fixes the admissible chamber, not the physical point (the  $D_4$ -fixed locus is positive-dimensional, U3); the discrete selectors  $\det R = 8$ ,  $\text{SNF}(R) = (1, 1, 8)$ ,  $\text{Spec}(Q_+) = \{1, 2, 3\}$  do the cutting. The gate splits into four pieces, three of them essentially closed:

- $U_{\text{unitary}}$  [N]: parabolic degree 0 + polystable  $\Rightarrow$  unitary (Mehta–Seshadri)  $\Rightarrow \Phi = 0$ ; the harmonic metric is the *finite* invariant form  $H$  ( $M_k^\dagger H M_k = H$ ), not a Hitchin PDE ([verification/v40\_harmonic\_metric.py]).
- $U_{\text{H2}}$  [N]:  $R, Q, K$  are read off the bundle (splitting type =  $a$ ,  $\det R = 8$ ,  $\text{Spec}(Q_+)$ ) ([verification/v39\_uwall\_selectors.py]).
- $U_{\Lambda^2}$  [I]: **Readout Rigidity** — on the discrete stratum  $\mathcal{S}_{8,Q} = \{\det R = 8, \text{SNF}(R) = (1, 1, 8), \text{Spec}(Q_+) = \{1, 2, 3\}\}$  the anchor-plane readout is *constant*,  $\mathcal{C}_{ud}(\rho) = g_{\text{car}} \|\text{Pl}_{1,a}(K)\|_1 / (N_{\text{fam}}^2 \Delta Q) = \frac{55}{117}$ , *independent* of the continuous  $D_4$  position — so  $c_u/c_d$  needs no point uniqueness ([verification/v49\_readout\_rigidity.py], [verification/v42\_exterior\_leg.py]).
- $U_{\text{point}}$  [A]: full uniqueness of  $\rho^*$  — needed only for the *complete*  $U_f^*$  amplitude matrix, *not* for  $c_u/c_d$ .

**Headline:** the remaining flavor bridge is not a PDE gate — it is a finite exterior *readout-rigidity* theorem; only  $U_{\text{point}}$  stays open.

**The simple closure** ([verification/v71\_simple\_r\_bridge.py]). The selector stratum  $\mathcal{S}_{8,Q}$  is now *fully derived*:  $\det R = 8 = \text{rank } E_8$  with  $\text{SNF}(R) = (1, 1, 8)$  is the lattice invariant (cf.  $\det Q = 3 = N_{\text{fam}}$ , v70), and  $\text{Spec}(Q_+) = \{1, 2, 3\}$  is the  $D_4$ -equivariant result (v69). The four flavor operators are integer lattice maps with compiler-atom determinants  $(\det Q, \det K, \det R, \det L) = (3, 4, 8, 20)$ , product  $1920 = |W(D_5)|$ . So by Readout Rigidity the quark *ratios* ( $c_u/c_d = g_{\text{car}} \cdot 11 / (N_{\text{fam}}^2 \cdot 13) = \frac{55}{117}$ , etc.) are *pure integer Plücker readouts* —

*no transcendental monodromy solve.* The earlier monodromy machinery was over-engineering for the ratios; the only transcendental piece,  $U_{\text{point}}$ , is the *absolute* amplitude normalisation — an anchor of the same nature as the one dimensional scale, not a missing computation.

**Gate 1 complete:**  $U_{\text{point}} \rightarrow v_{\text{geo}}$  (`[verification/v75_upoint_to_vgeo.py]`). The absolute amplitudes are not free either. The lepton amplitudes are exact ( $c = (\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$ , v20), every cross-sector  $c$ -ratio is closed (Plücker), and each *sector product* is a clean  $\varphi_0^{\text{ret}}$ -power (Grand Mass Volume, v46:  $\det M_{\text{sector}} \sim (\varphi_0^{\text{ret}})^{6,9,10}$ ; the lepton product is  $\frac{16}{7} \cdot \frac{4}{3} \cdot \frac{7}{6} = \frac{32}{9} = 2^{g_{\text{car}}} / N_{\text{fam}}^2$ ). Since (pairwise ratios) + (product)  $\Leftrightarrow$  (individuals) is a bijection, the nine charged-fermion amplitudes are *fixed up to one overall scale*  $v_{\text{geo}}$ . **So  $U_{\text{point}}$  is not an independent gate: it reduces to  $v_{\text{geo}}$ , the same one irreducible dimensionful anchor as gravity's  $1/G$  (v68) — the two  $[A]$  anchors collapse to one.**

## Research Contract 2 — ( $G_{\text{metric}}$ )

**Goal.** Construct the TFPT metric measure

$$Z[J] = \lim_{\chi \rightarrow \infty} \int_{\mathcal{G}_\chi / \text{Diff}} \exp(-S_{\text{rel},\chi}[g, A, \psi] + J \cdot O) d\mu_{\text{Cal},\chi},$$

$$S_{\text{rel},\chi} = \text{Tr} f\left(\frac{D_{\text{rel}} + A_\Sigma}{\chi}\right) - \text{Tr} f\left(\frac{D_{\text{ref}}}{\chi}\right) + \frac{i\pi}{2} \Delta\eta_\Sigma,$$

the relative spectral action whose low-curvature shadow is the already-closed  $R + R^2$  equation.

**Lemma 8** (G0–G1 — configuration space & BRST/Hodge Fredholm). *For  $s > 5/2$  a Sobolev space of  $\Theta$ -symmetric metrics on the doubled seam collar  $M_\Sigma$  admits a gauge slice; the gauge-fixed gravitational complex  $\Omega^0(TM) \xrightarrow{K} \Gamma(S^2 T^*M) \xrightarrow{K^*} \Omega^0(TM)$  with  $\Delta_{\text{gh}} = K^*K$  and  $\Pi_{\text{phys}} = \Pi_{\text{BRST/Hodge}} \Pi_{TT}$  is elliptic Fredholm; the conformal mode is removed as BRST-exact / non-propagating in the Calderón polarisation (not rhetorically). Machine: principal symbols in xAct/-Cadabra; linear-algebra certificates in Lean (medium).*

**Lemma 9** (G2 — finite relative spectral action, heat-kernel grounded [N]).  *$S_{\text{rel},\chi}$  is finite, differentiable, with a local Seeley–DeWitt expansion  $\text{Tr} f(D^2/\chi^2) \sim f_4 \chi^4 a_0 + f_2 \chi^2 a_2 + f_0 a_4 + \dots$ . For the Lichnerowicz Dirac operator  $D^2 = \nabla^2 + \frac{R}{4}$  (so  $E = -\frac{R}{4}$ , spinor trace  $\text{tr} \mathbf{1} = 4$ ) the Gilkey coefficients give, per  $(4\pi)^{-2}$ ,*

$$a_2 = \frac{1}{6} \text{tr}(6E + R \mathbf{1}) = -\frac{R}{3} \quad (\text{Einstein–Hilbert } \int R), \quad a_4|_{R^2} = \frac{1}{360} \text{tr}(180E^2 + 60RE + 5R^2 \mathbf{1}) = \frac{R^2}{72}$$

(the  $R^2$ /Starobinsky term), so the spectral action structurally produces  $S_{\text{eff}} = \int \sqrt{g} \frac{\bar{M}_{\text{Pl}}^2}{2} (R + \frac{R^2}{6\bar{M}^2}) + \dots$  with the scalaron mass ratio fixed by the cutoff moment,  $M^2/\bar{M}_{\text{Pl}}^2 = 6(4\pi)^2/f_0$ . The TFPT closure  $M^2/\bar{M}_{\text{Pl}}^2 = c_3^7$  fixes  $f_0 = 6(4\pi)^2/c_3^7$ , i.e.  $M = c_3^{7/2} \bar{M}_{\text{Pl}} = 3.06 \times 10^{13} \text{ GeV}$  — the boundary normalisation  $c_3$  is the gravitational scale — and hence the Starobinsky slow-roll forms  $n_s \simeq 1 - \frac{2}{N}$ ,  $r \simeq \frac{12}{N^2}$ . The  $R + R^2$  structure is convention-independent (standard Gilkey); the precise rational  $\frac{1}{72}$  and factor 6 are the standard Dirac conventions used here (`[verification/v36_spectral_action_g2.py]`, `[verification/v28_gravity_fR.py]`).

**The simpler resolution: “full QG” for physics is the gap-decoupled admissible sector [F]**

The infinite-dimensional ambient measure (G6) is the summit, but the admissible IR sector does not wait for it. The metric coupling is gap-dominated (G5):  $2\|V_{\text{metric}}\| = 0.785 < \Delta = 6 \log \frac{3}{2} = 2.433$ , so the admissible sector keeps a *strictly positive* effective gap  $\Delta_{\text{eff}} = 1.648 > 0$  under the stated RP and metric-coupling assumptions. A positive gap dynamically *isolates*

the admissible sector from the un-built ambient/deep-UV: under those assumptions the admissible (IR) measure is closed, and the ambient completion is decoupled from it (we do *not* assert it is physically irrelevant — that would itself be a physical interpretation). So the honest reading is

$$\begin{aligned} \text{admissible IR sector} &= \text{closed under stated RP}; \\ \text{ambient metric measure (G6)} &= G_{\text{metric gate}} \end{aligned}$$

G2 supplies the local action ( $R + R^2$ , heat-kernel); G5 supplies the decoupling of the admissible sector; together they reduce ( $G_{\text{metric}}$ ) to the ambient projective limit (G6). **Precise status (Alessandro’s boundary):** the admissible IR sector is closed under the stated RP and gap-dominance assumptions; the ambient metric projective measure remains a *strict TOE completion target*. Its absence does *not* affect the bounded IR claim, but it *does block certification as a strict physical TOE* — so we do not call it “mere mathematical completeness”. [verification/v36\_spectral\_action\_g2.py]

**The glue Q-system: the index-4 object of the  $G_{\text{net}}$  statement exists explicitly [I]/[P] ( [verification/v125\_glue\_qsystem.py] )**

The  $G_{\text{net}}$  target — “the seam–Calderón inclusion has Jones index  $4 = |\mu_4|$ ” — no longer refers to an unknown object. On the glue group algebra  $A = \mathbb{C}[\mathbb{Z}_4]$  (multiplication  $\delta_a \otimes \delta_b \mapsto \delta_{a+b}$ , unit  $\delta_0$ ) *all* Longo Q-system axioms hold exactly [I]: associativity, unit law, the Frobenius identity  $(m \otimes 1)(1 \otimes m^*) = m^*m = (1 \otimes m)(m^* \otimes 1)$ , and specialness  $mm^* = |\mathbb{Z}_4| \mathbf{1}$  — a special  $C^*$  Frobenius algebra with

$$\dim \theta = 4 = |\mu_4| = [B : A]$$

exactly the Carrier Index Lemma value (KLM:  $\mu_{\text{carrier}} = 16 = 4^2 \mu_{E_8}$  re-verified). *Locality = isotropy* [I]: on the Lagrangian glue  $\langle (1, 1) \rangle$  the discriminant form gives  $q(k(1, 1)) = k^2 \in \mathbb{Z}$  for every  $k$  — the braiding obstruction vanishes (v92 in Q-system language); the two Lagrangian glues (the sheet pair) are the only maximal isotropic choices. *Halfway step* [I]:  $\{0, 2\} < \mathbb{Z}_4$  closes and gives the unique intermediate Q-system  $\mathbb{C}[\mathbb{Z}_2]$  = the  $SO(16)_1$  tower step, with index factorisation  $4 = 2 \times 2$ . *R2 sharpened* [P] (recorded, not claimed): what remains is the operator-algebraic *identification* “the seam–Calderón inclusion of the 16-Majorana net (v113) realises *this* Q-system” — matching two constructed objects, not a search; the three loads (metric + carrier + QBL, v123) hang on this one identification.

**The hull is the graded module of the Q-system ( [verification/v128\_graded\_hull.py] ).** The Q-system is not merely an abstract companion of  $E_8$  —  $E_8$  *realises* it [I]: the 240 roots, built as explicit norm-2 vectors in  $D_5 \oplus A_3$ +glue coordinates, decompose as  $240 = 52 + 64 + 60 + 64$  over the glue  $\mathbb{Z}_4$  (the v89 branching as vectors), and for *all* 6720 root pairs whose sum is again a root,  $\text{coset}(r_1 + r_2) = \text{coset}(r_1) + \text{coset}(r_2) \bmod 4$ :

$$E_8 \text{ is a } \mathbb{Z}_4\text{-graded Lie algebra over the carrier; the grading group is the glue = the Q-system}$$

— the checksum hull is the Q-system’s graded module, realised in 248 dimensions. The R2 identification thus matches the Calderón inclusion against a structure that  $E_8$  already *carries*.

**Lemma 10** (G3–G4 — reflection positivity, fixed and integrated). *For each  $\Theta$ -invariant  $g$  in the Calderón ball the physical boundary kernel  $\mathcal{K}_g = P_{\text{adm}} \Pi_{\text{phys}} \mathcal{N}_g^{TT} \Pi_{\text{phys}} P_{\text{adm}}$  is reflection positive,  $\langle \Theta F, F \rangle_{\mathcal{K}_g} \geq 0$  (G3); and  $d\mu_{\text{Cal}, \chi}$  is  $\Theta$ -invariant, supported where G3 holds uniformly, so  $\int \langle \Theta F, F \rangle_{\mathcal{K}_g} d\mu_{\text{Cal}, \chi} \geq 0$  (G4). **G4 is the decisive step:** RP after integrating over dynamical*

metrics, not merely at fixed background.

**Lemma 11** (G5 — gap dominance [F]).  $2\|V_{\text{metric}}\|_{\text{rel}} < \Delta = 6 \log \frac{3}{2}$ , with  $\|V_{\text{metric}}\|_{\text{rel}} \leq 248c_3^2 = \frac{31}{8\pi^2} = 0.39262$ , hence  $2\|V\| = 0.7852 < 2.4328$  and  $\Delta_{\text{eff}} \geq \Delta - 2\|V\| > 1.647 > 0$ . The numerical inequality is trivial ([verification/v29\_research\_contract\_certs.py]); the work is the operator-norm bound (hard).

**Theorem 1** (Decoupling theorem — physical low-energy claims do *not* need the full  $G_{\text{metric}}$ ). If the admissible sector is reflection-positive (G4) and the metric coupling is gap-dominated,  $2\|V_{\text{metric}}\| < \Delta = 6 \log \frac{3}{2}$  (G5), then the admissible IR measure is stable under the ambient completion: a strictly positive effective gap  $\Delta_{\text{eff}} = \Delta - 2\|V\| > 0$  protects the admissible sector from the (un-built) ambient/deep-UV, so every physical low-energy read-out (masses, mixings,  $\alpha^{-1}$ , the  $R+R^2$  regime) is independent of the open ambient measure. Symbolically

$$\boxed{\text{RP} + 2\|V\| < \Delta \implies \text{admissible IR measure stable under ambient completion .}}$$

Thus ( $G_{\text{metric}}$ ) is not a diffuse “full quantum gravity missing” block: it is the sharply delimited ambient projective limit (G6), a strict-TOE certification target whose closure the falsifiable IR physics does not depend on. **Typing:** the inequality is machine-checked [F]; the operator-norm bound on  $\|V_{\text{metric}}\|$  is the conditional input.

**Lemma 12** (G6–G7 — projective limit & Ward identities). A projective system  $\mu_{\chi_n}$  with  $\pi_{m,n*}\mu_{\chi_m} = \mu_{\chi_n}$ , uniform moment bounds  $\sup_n \int \|g\|_{H^s}^p d\mu_{\chi_n} < \infty$ , tightness and regulator-independence yields  $\mu_{\text{QG}} = \lim \mu_{\chi}$  on  $\mathcal{G}/\text{Diff}$  (G6, the hardest step); the limit obeys the BRST Ward identities  $\langle s\mathcal{O} \rangle = 0$ , whence  $\nabla^\mu T_{\mu\nu} = 0$  and the  $\eta$  boundary phase cancels the APS anomalies (G7).

**Gate 2 reduction:** G6 is a *boundary* projective limit, not a bulk one ([verification/v76\_gmetric\_reduction.py])

The exact decoupling margin is  $\Delta_{\text{eff}} = \Delta - 2\|V\| = 6 \log \frac{3}{2} - \frac{31}{4\pi^2} = 1.648 > 0$ , with  $\|V_{\text{metric}}\|_{\text{rel}} \leq 248 c_3^2 = \dim(E_8) c_3^2 = \frac{31}{8\pi^2}$  ( $31 = 2^{g_{\text{car}}} - 1$ ). **Holographic reduction:** because the seam is a *finite causal boundary*, the ambient *bulk* metric measure is reconstructed from a finite-codimension seam (Calderón) *boundary* measure — so G6 is a *boundary* projective limit. The conditional theorem is then

$$\boxed{\text{RP}(\text{seam-Calderón boundary kernel}) + \text{tightness} \implies G_{\text{metric}} \text{ closes}}$$

i.e. the dimension of the open problem drops from bulk QG to a finite seam-boundary measure. **Residual:** the constructive boundary projective limit (constructive-QFT grade) — reduced, not closed. So Gate 2 is now *two clean tiers*: Tier A (IR physics) closed under RP+gap; Tier B (strict TOE) reduced to a boundary measure, no longer a diffuse “full QG missing”. [P]/[A]

**And that boundary measure is a *rigorously constructed net*** ([verification/v77\_e8\_conformal\_net.py]). The level-1 central charges  $c = \dim \mathfrak{g}/(1 + h^\vee)$  give  $c(E_8)_1 = \frac{248}{31} = 8 = \text{rank } E_8$ ,  $c(D_5)_1 = \frac{45}{9} = 5$ ,  $c(A_3)_1 = \frac{15}{5} = 3$ , and  $c(D_5) + c(A_3) = 5 + 3 = 8 = c(E_8)$  is a *conformal embedding*  $(D_5)_1 \times (A_3)_1 \subset (E_8)_1$  (coset  $c = 0$ ). Now  $(E_8)_1$  is the holomorphic  $c = 8$   $E_8$ -lattice VOA / conformal net — one of the most rigorously constructed objects in mathematical QFT (Frenkel–Kac–Segal; Kawahigashi–Longo / Carpi). **So the conditional theorem becomes concrete:** *identify the seam-Calderón boundary measure with the  $(E_8)_1$  lattice net* (carrier =  $(D_5)_1 \times (A_3)_1$  subnet), and the gap  $\Delta_{\text{eff}} > 0$  supplies the clustering/tightness — so G6 is imported into existing RCFT/conformal-net rigor rather than built from scratch. Residual = the net identification + bulk reconstruction. [I] (charges, embedding) + [P] (identification)

### Theorem G — the closure statement [A] (target)

The relative spectral boundary kernel over the diffeomorphism-quotiented metric sector admits a regulator-independent, reflection-positive projective-limit measure  $\mu_{\text{QG}}$ . Its Ward identities imply covariant conservation, and its low-curvature expansion is the TFPT  $R + R^2$  seam action

$$f_R R_{\mu\nu} - \frac{1}{2} f(R) g_{\mu\nu} + (g_{\mu\nu} \square - \nabla_\mu \nabla_\nu) f_R = \bar{M}_{\text{Pl}}^{-2} T_{\mu\nu} + O(R^3/M^4),$$

so the closed covariant equation becomes the controlled IR shadow of the full measure. *Status: open; the deepest item. G5 certified; the regime equation closed [verification/v28\_gravity\_fR.py].*

## Certifiability and recommended order

| Building block                                | Machine                 | Hardness             |
|-----------------------------------------------|-------------------------|----------------------|
| $W_{\text{wall}}$ enumeration ( $C_U^{(1)}$ ) | Lean / Python / Wolfram | easy ( <i>done</i> ) |
| $D_4$ fixed-locus polynomial ( $C_U^{(2)}$ )  | Sage / Mathematica      | medium               |
| selector uniqueness ( $C_U^{(3)}$ )           | interval arithmetic     | medium               |
| SNF, determinants, spectra                    | Lean                    | easy-medium          |
| principal-symbol ellipticity (G1)             | xAct / Cadabra / Lean   | medium               |
| gap inequality (G5)                           | Lean                    | easy ( <i>done</i> ) |
| operator-norm bound (G5)                      | analysis + numerics     | hard                 |
| projective limit (G6)                         | not primarily machine   | very hard            |
| Ward identities (G7)                          | BV/BRST formal + checks | hard                 |

### Recommended order

$$(U_{\text{wall}}) \rightarrow \text{freeze the frontier status} \rightarrow (G_{\text{metric}})$$

$(U_{\text{wall}})$  can visibly upgrade the theory — it hits the last concrete flavor ambiguity, including the “11” — and it is finite and falsifiable (the U2 kill switch can destroy it cheaply).  $(G_{\text{metric}})$  is more fundamental but a *programme*, not a single proof: heavy machinery, with the projective limit (G6) the genuine summit.